

JEE MAIN REVISION MASTER: ISOMERISM

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 10 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Basics Of Organic Chemistry — Core Principles & Governing Laws	5 PYQs	Core Principle
Moderate	Basics Of Organic Chemistry — Applied Cases, Reactions & Traps	5 PYQs	Applied Topics
Standard	Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Basics Of Organic Chemistry — Core Principles & Governing Laws	$[\text{Fe}(\text{O}-\text{NO})_3\text{Cl}_3]$, $\text{CH}_3\text{CH}_2\text{CHO}$, C_5H_{12}
Basics Of Organic Chemistry — Applied Cases, Reactions & Traps	$\text{CH}_3\text{CH}=\text{CHCH}_3$, $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_3$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Basics Of Organic Chemistry — Core Principles & Governing Laws (Moderate Yield — 5 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Basics Of Organic Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Basics Of Organic Chemistry PYQs.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The number of chain isomers possible for the hydrocarbon with molecular formula $\text{C}_{(5)}\text{H}_{(12)}$ is..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Propanone and propanal are..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "Which of the following is not a pair of functional isomers?..." [Ans: N/A]

3.2 Basics Of Organic Chemistry — Applied Cases, Reactions & Traps (Moderate Yield — 5 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "Among the following, identify the compound that is not an isomer of hexane..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "A pair of compounds having the same boiling points are..." [Ans: N/A]
- [2020 Shift] Q#1 (MCQ): "In the reaction, The number of possible isomers for the organic compound X is..." [Ans: N/A]

3.3 Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The number of chain isomers possible for the hydrocarbon with molecular formula $C_{(5)}H_{(12)}$ is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Among the following, identify the compound that is not an isomer of hexane..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The number of chain isomers possible for the hydrocarbon with molecular formula $C_{(5)}H_{(12)}$ is

- (A) 4
- (B) 3
- (C) 2
- (D) 1

Correct Answer: None

Solution Analysis: For the hydrocarbon with molecular formula $C_{(5)}H_{(12)}$, we need to count the possible chain isomers. $C_{(5)}H_{(12)}$ is an alkane. Chain isomerism occurs when the carbon skeleton is arranged in different ways. The possible structures are: Straight chain: $CH_3-CH_2-CH_2-CH_2-CH_3$ This is pentane. One methyl branch: $CH_3-CH(CH_3)-CH_2-CH_3$ This is 2-methylbutane. Two methyl branches on the middle carbon: $C(CH_3)_4$ This is 2,2-dimethylpropane. So, the

Benchmark Problem 2: KCET 2023 (Q#4)

A pair of compounds having the same boiling points are

- (A) cis but-2-ene and trans but-2-ene
- (B) n-hexane and neo-hexane
- (C) benzene and naphthalene
- (D) (+) butan-2-ol and (-) butan-2-ol

Correct Answer: None

Solution Analysis: The correct answer to the question: "A pair of compounds having the same boiling points are," is Option D: (+) butan-2-ol and (-) butan-2-ol. Let's explain why the other options are incorrect: Option A: cis but-2-ene and trans but-2-ene have different spatial arrangements around the double bond. This difference in geometry leads to different physical properties, especially boiling points. Typically, the trans isomers are less polar and have a lower boiling point compared to the cis isomers. Option B: n-hexane and neo-hexane (more correctly called 2,2-dimethylbutane) are structural isomers,